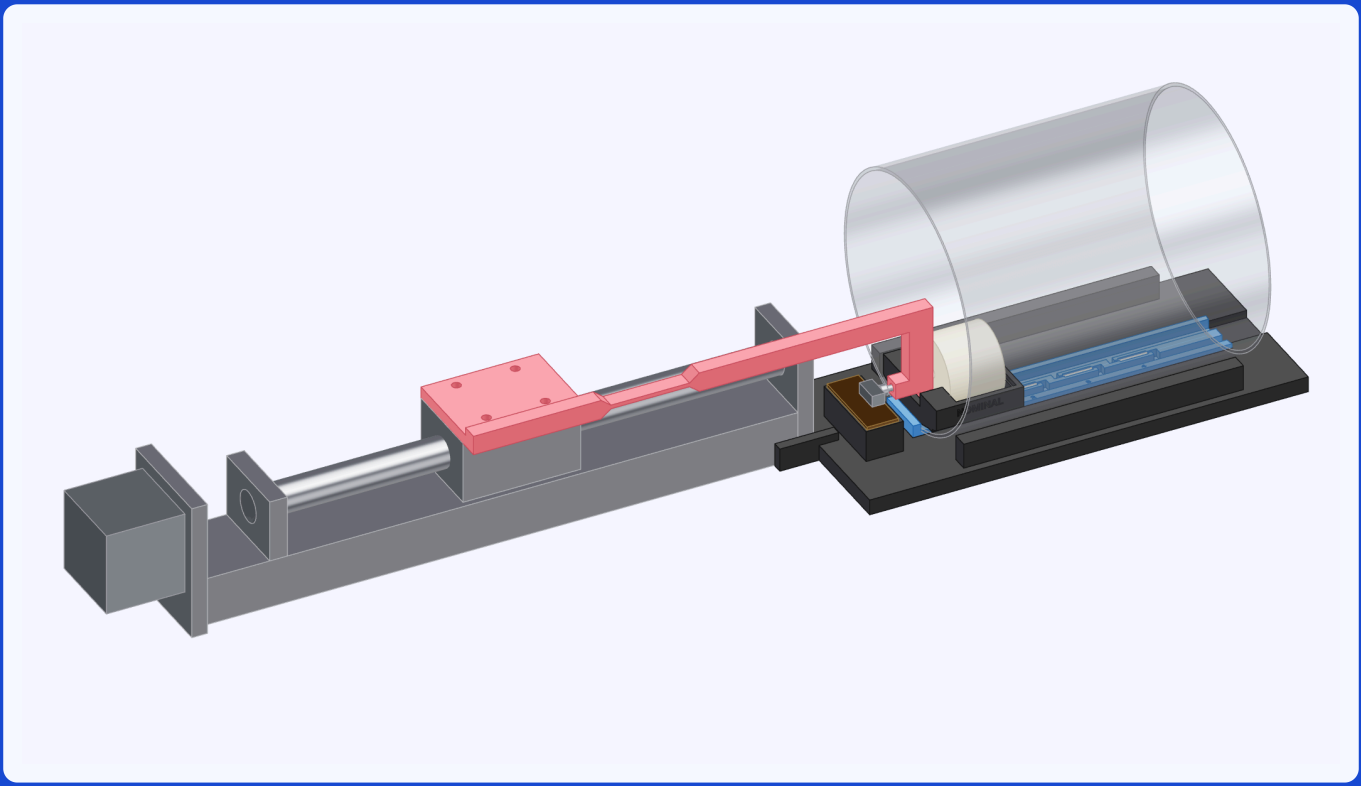
Reed sweep rig

Build the bench fixture that moves the Aero float past the carbonator's level reeds and records where each contact closes and opens.



BUILD ORDER

The whole build

Print the parts, assemble the dry rig, then prove the wiring and motion before taking calibration readings.

OPERATION	PAGES
Before the bench The fixture, bench kit and print plates	3–5
Assemble the rig Inserts, switch, push rod, reed comb, tube and slide	6–10
Set the float and home Sled, roller, clearances and striker	11–14
Wire the controller Driver, supplies, motor, switch and reeds	15–17
Check the wiring Meter checks, USB inputs and drive current	18–20
First motion and calibration Jog, home, record a sweep and use the directional crossings	21–23

HOW TO READ THE PAGES

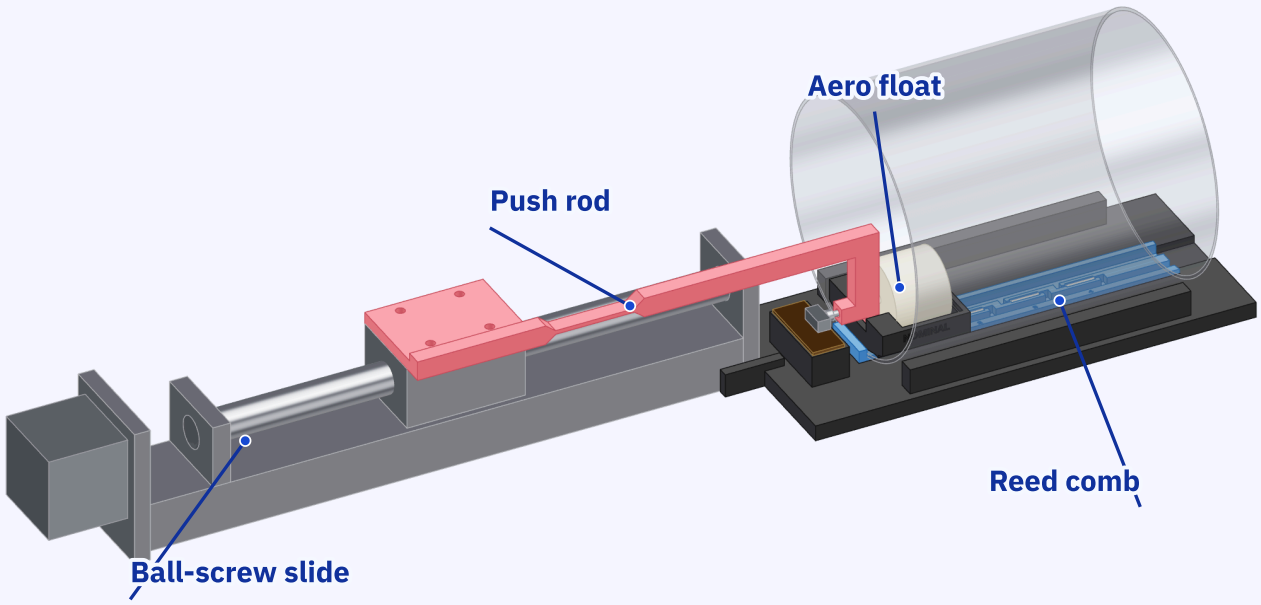
Coral marks the part in hand or the direction it moves. Brass marks a heat-set insert. Blue marks a datum or gauge. The tube is translucent where you need to see inside. The boxed check tells you when to continue.

Enter a page number in the website reader to jump to an operation. A PDF app can use the bookmarks. Print single-sided on Letter at actual size.

BEFORE THE BENCH

How the rig works

The slide drives the float lengthwise. The reeds sit under the tube wall on the same line the float passes over.



The rim at the rod end seats on the comb's tab. The reader uses that rim as the carbonator's bottom reference, with positions increasing into the tube.

The sled sets the gap

NEAR, **NOMINAL** and **FAR** place the same float progressively farther from the reeds. The sled's runners ride on the bore.

The switch sets home

The striker trips the switch as the sled moves toward the mouth. The distance from that trip to the tab ties the step count to the tube.

SET UP ON WOOD OR PLASTIC

Keep loose steel, rulers and tools a hand's width from the tube during readings. Use the specified stainless fasteners by the mouth. The slide stays out at the end of the long push rod.

BEFORE THE BENCH

What to have on the bench

Lay out the motion parts, controller and test pieces before heating the insert tool.

WHERE IT GOES	PART
Slide	Befenybay BE069-2 ball-screw stage with its NEMA 17 motor and lead
Drive and power	BIGTREETECH TMC2209 V1.3, ZXXP stepper driver carrier and BTF-LIGHTING 12 V adapter with its female barrel connector
Logic	DORHEA ESP32 with screw-terminal breakout and a USB-C data cable
Home	R REIFENG endstop board and lead
Test pieces	Bare carbonator tube, finished Aero float, and the reeds you intend to install
Fasteners	Three ruthex M3 heat-set inserts, two 316 stainless M3 × 8 switch screws, one 304 stainless M3 × 12 striker, four M4 carriage screws and washers
Wire	Thin insulated signal and return wires for the comb, plus Dupont jumpers for the carrier

Bench tools

Heat-set tip, soldering iron, fine pliers, cutters, hex keys, multimeter, calipers and smooth gauge pins or drill shanks. Set out a bank card for the striker and an insulated tool for the driver's trimmer.

Tool ledger and purchased part links

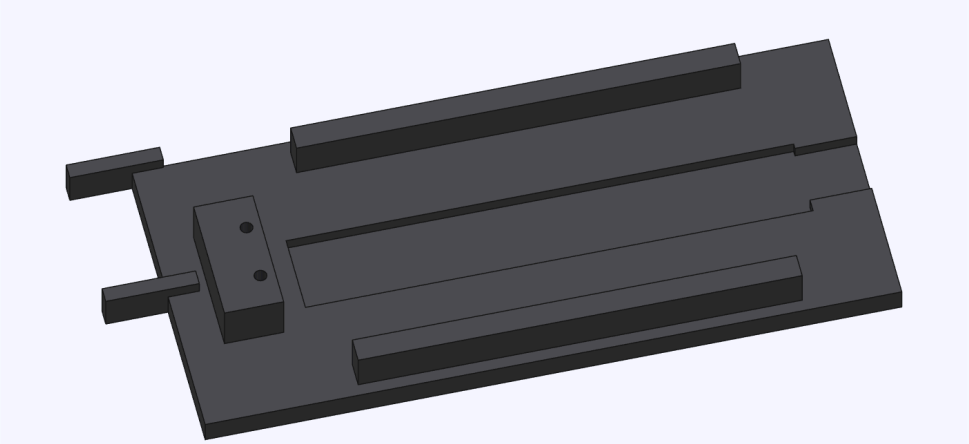
BEFORE ASSEMBLY

Dry-fit the bought slide and switch against the printed cradle. Their modeled dimensions come from supplier drawings. Check the actual hole pattern and seating faces before setting inserts or forcing a fit.

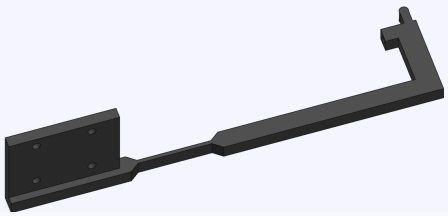
PRINT

Print the cradle and small parts

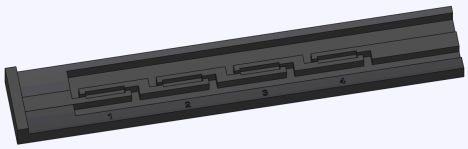
Choose both prepared plates for the same printer. Use black PET-GF, with the orientations and support settings already in the projects.



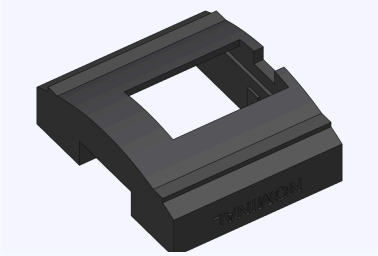
Cradle plate. The cradle lies flat, with the comb pocket, switch pad and keepers up.



Push rod. The pad stands on edge.



Reed comb. Pockets and wire grooves face up.



Sleds. Print NEAR, NOMINAL and FAR upside down, with the runners up.

Small-parts plate

The rod, comb and all three sleds share one plate. Both prepared plates print without supports.

Mark1 and Mark2 print files

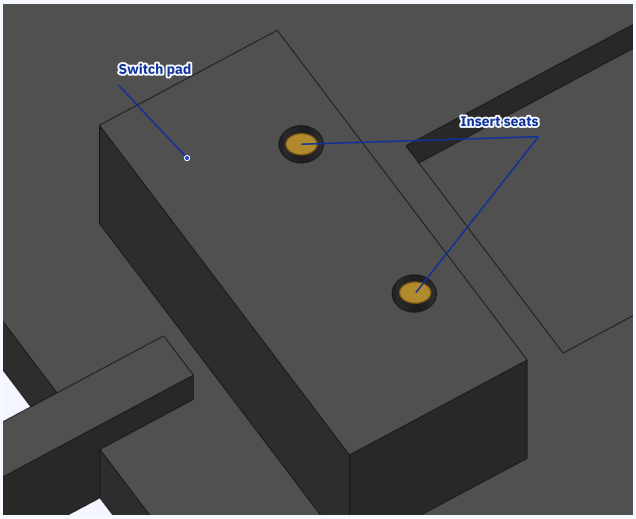
BEFORE HEAT-SETTING

The comb drops fully into its pocket. Clear stray strands from the reed pockets, sled slots and gauge windows. Keep the sleds' runner edges square.

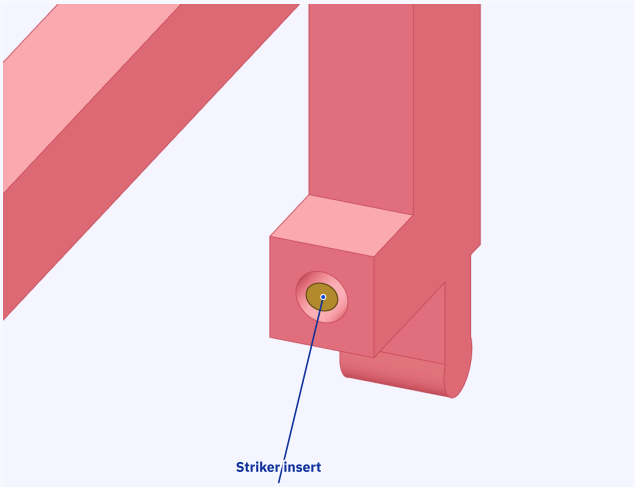
ASSEMBLE THE RIG

Heat-set the inserts

The cradle's switch pad gets a pair of inserts. The third goes in the end face of the push rod's finger.



The holes on the raised pad take the switch screws.



Stand the rod so you can press squarely into the finger.

1. Check the endstop board's mounting holes against the pair on the pad. Clear stray filament from all insert holes.
2. Use the heat-set tip to press each ruthex M3 insert in on its own axis. Stop flush with the surrounding face.
3. Let the plastic cool. Hand-start the matching screw in each insert.

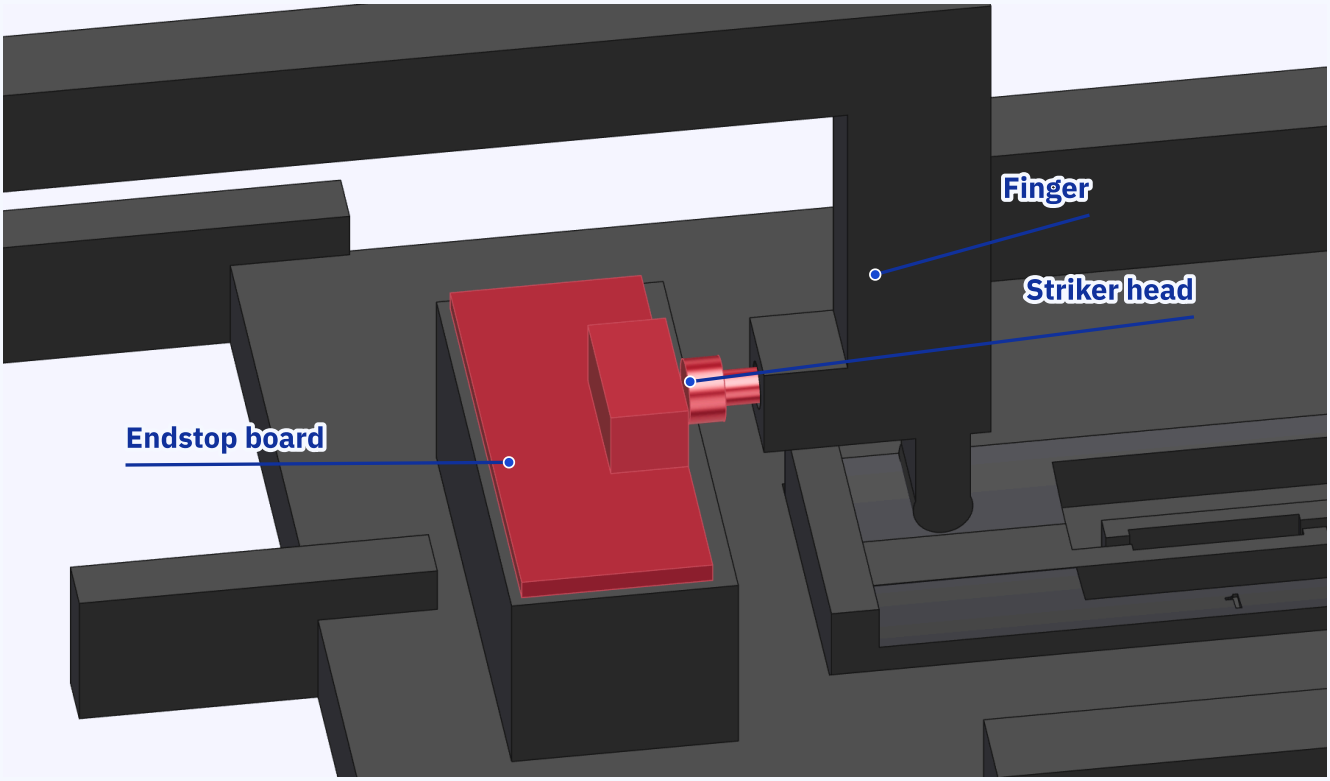
CHECK THE SEATING FACES

The switch board can lie flat and each screw threads by hand. Clear any raised plastic around the insert before fitting the board.

ASSEMBLE THE RIG

Fit the switch and striker

The switch faces the tube. The striker head faces back toward the switch as the sled approaches the mouth.



The endstop mounts through the holes beside the switch. The CAD shows the switch body in place. Use the lever on the actual board for the striker check.

1. Set the board on the cradle pad with the lever toward the tube. Fasten it with the 316 stainless M3 × 8 screws.
2. Hand-thread the 304 stainless M3 × 12 into the finger, head toward the lever. Run it most of the way in. Final adjustment comes after the sled is coupled.

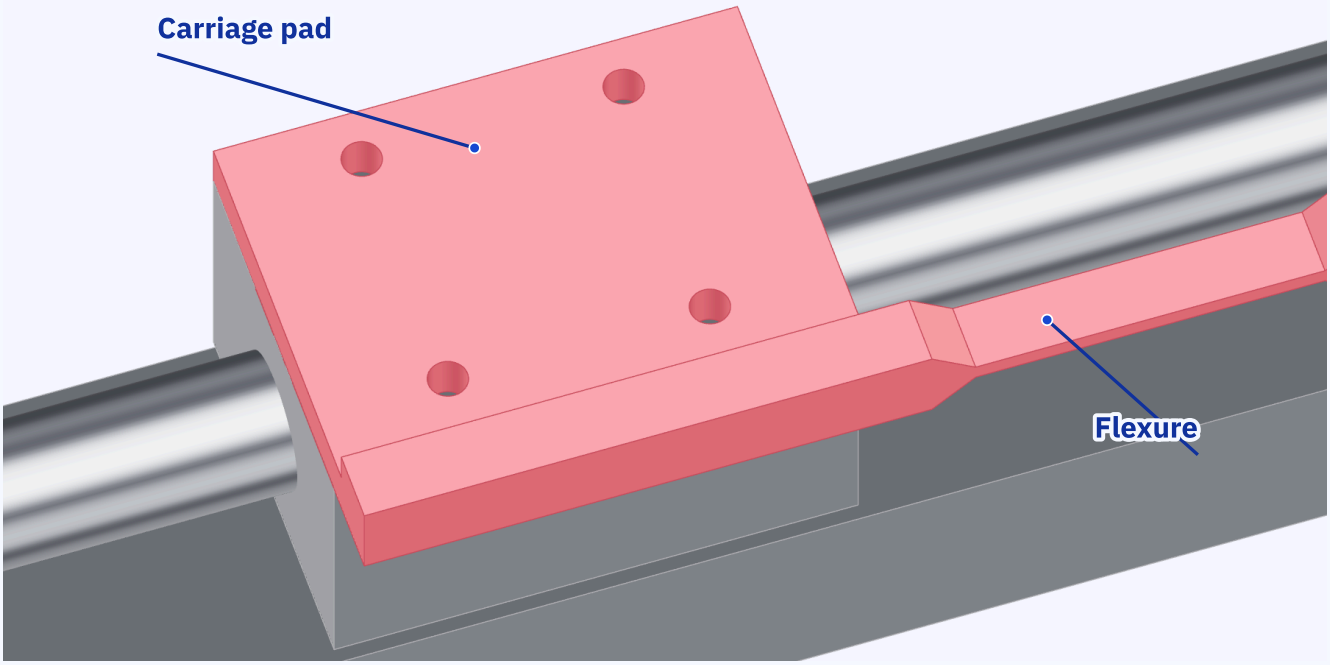
CHECK THE SWITCH BY HAND

The board sits flat and stays put. The lever moves freely and returns when you release it.

ASSEMBLE THE RIG

Bolt the rod to the carriage

The rod pad sits on top. Its bar runs beside the carriage, with the roller leg pointing down at the tube end.



The shallow section beside the pad is the flexure. Lift the far end only enough to let the roller clear the sled.

1. With the stage unpowered, turn the coupling by hand until the carriage is near the middle.
2. Try an M4 screw by hand in a bare carriage hole. Compare its available thread depth with the screw's projection through the pad and washer. Add washers or use a shorter screw if it would bottom.
3. Set the pad flat on the carriage and fasten it through all four holes with the M4 screws and washers.

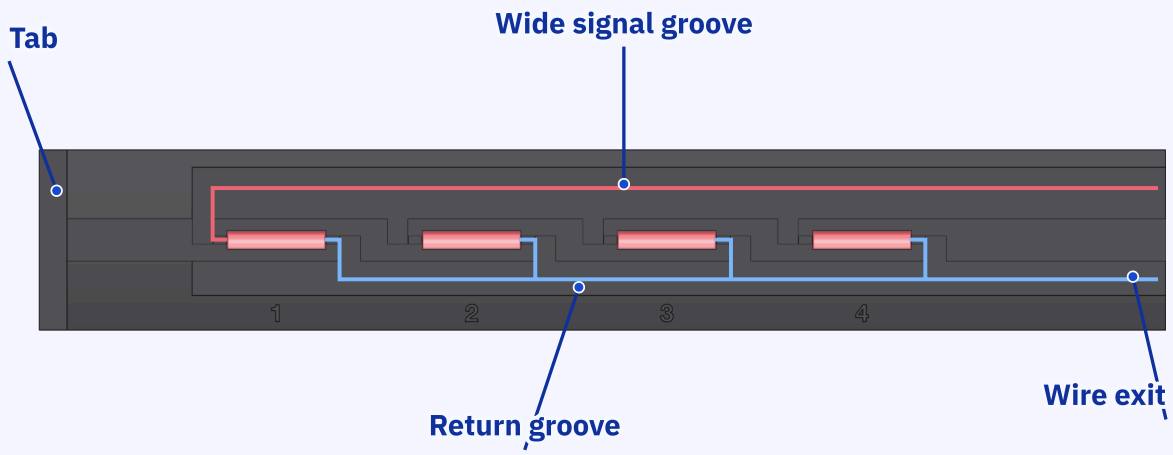
CHECK THE CARRIAGE

The pad is flat and cannot rock. The screws clamp the pad before bottoming. The flexure moves the roller and springs it back when released.

ASSEMBLE THE RIG

Wire the reed comb

Use the reeds intended for the machine. The tab end of each reed goes to its own signal, and the far lead joins the common return.



Stations count from the tab. The low and high positions are the middle pair. One signal route is shown in coral. Blue shows the shared return.

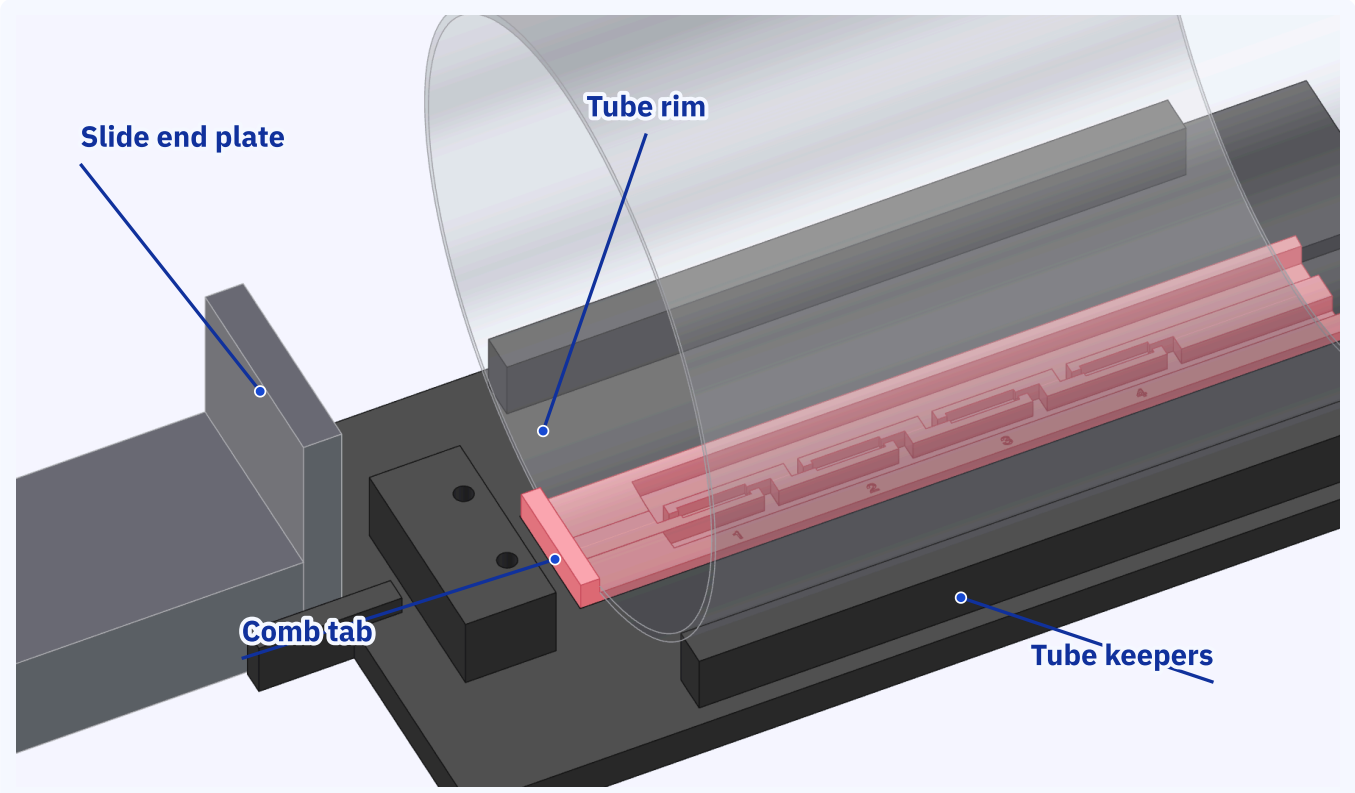
1. Place each glass body in its numbered pocket and seat it against the tab-side end.
2. Support each lead with pliers between the glass and the bend. Take the tab-side lead through its slit into the wide groove and solder its own thin, labeled signal wire.
3. Bend the other lead into the narrow groove and solder it to the common return. Trim each lead inside its groove and take the wires out the end opposite the tab.

CHECK CLEARANCE AND CONTINUITY

Glass, wire and solder sit below the land that carries the tube. Meter each signal against the return. Moving the float's magnet past that reed produces a closure and a release.

Seat the comb, tube and slide

Use a flat wood or plastic top so the cradle and slide stand at their intended heights.



The slide approaches the cradle from the left. The tube rim and sled share the tab as their reference.

1. Drop the comb into its pocket with the tab toward the switch pad. Lay the wires in the trough at the far end.
2. Lower the bare tube between the keepers with its weld seam up. Slide the rim against the tab.
3. Bring the slide's non-motor end between the cradle cheeks. Seat the end plate against the cradle and align the rod with the tube mouth.

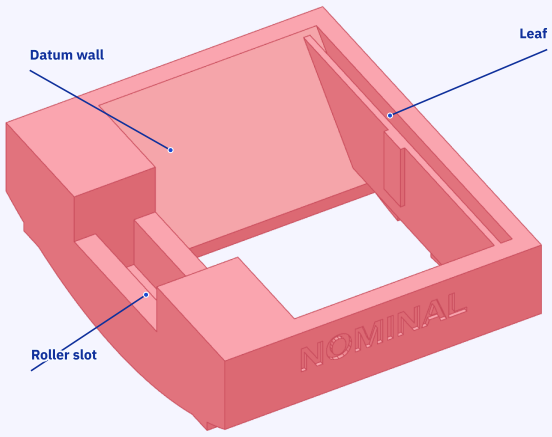
CHECK WHAT BEARS THE LOAD

The tube rests on the comb without rocking or pinching a wire. The rim touches the tab and the end plate sits squarely between the cheeks.

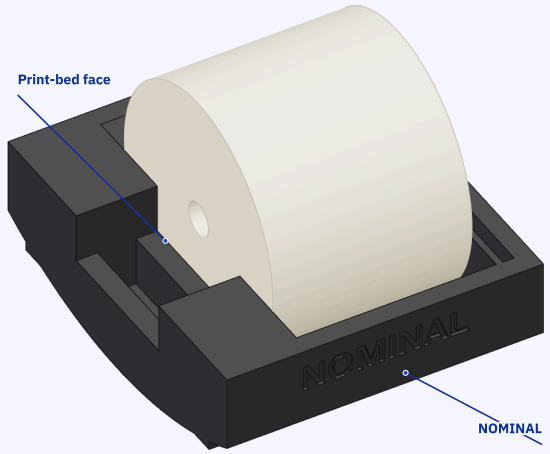
SET THE FLOAT AND HOME

Seat the float in the sled

Start with NOMINAL. The face that was on the print bed goes against the datum wall beside the roller slot.



Empty. The datum wall is between the slot and the vee.



Seated. The float lies on its side. The leaf holds its bottom face against the datum wall.

1. Find the float's print-bed face. Hold it flat against the datum wall, with the guide bore along the sled.
2. Press down into the vee until the far face passes the leaf and the leaf springs back.

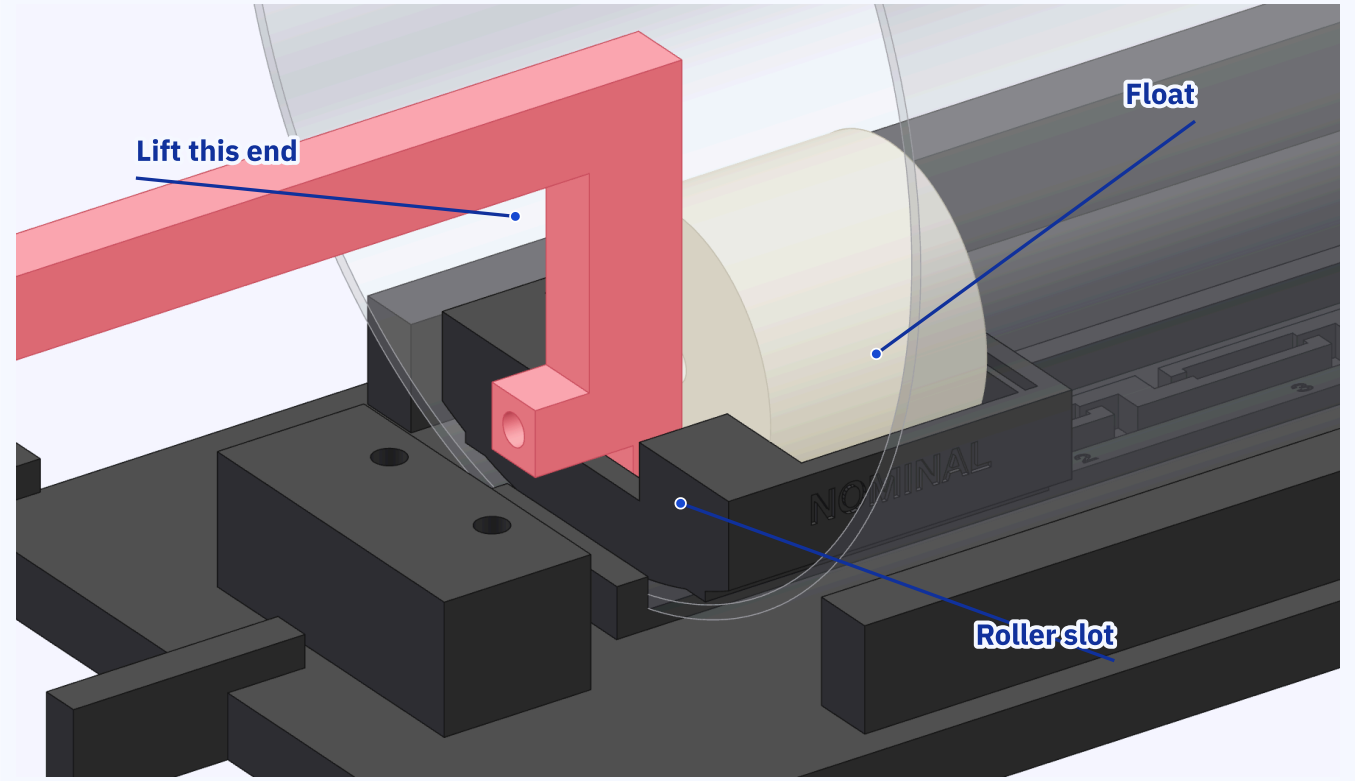
CHECK THE SEAT

The float bears on the vee and its bottom face stays against the datum wall. The roller slot remains clear. The float stays seated when you turn the sled in hand.

SET THE FLOAT AND HOME

Couple the sled to the rod

Put the loaded sled into the open end away from the slide, slot first. The roller drops in at the mouth.



The rod leg drops into the slot ahead of the float. The slot's front wall hides the roller once it seats.

1. With motor power disconnected, feed the sled toward the mouth until the slot is under the rod leg.
2. Lift the far end of the rod just enough to clear the sled, then lower the roller fully into the slot.
3. Turn the slide coupling by hand to move the sled away from the tab and back. Watch both runners and the roller.

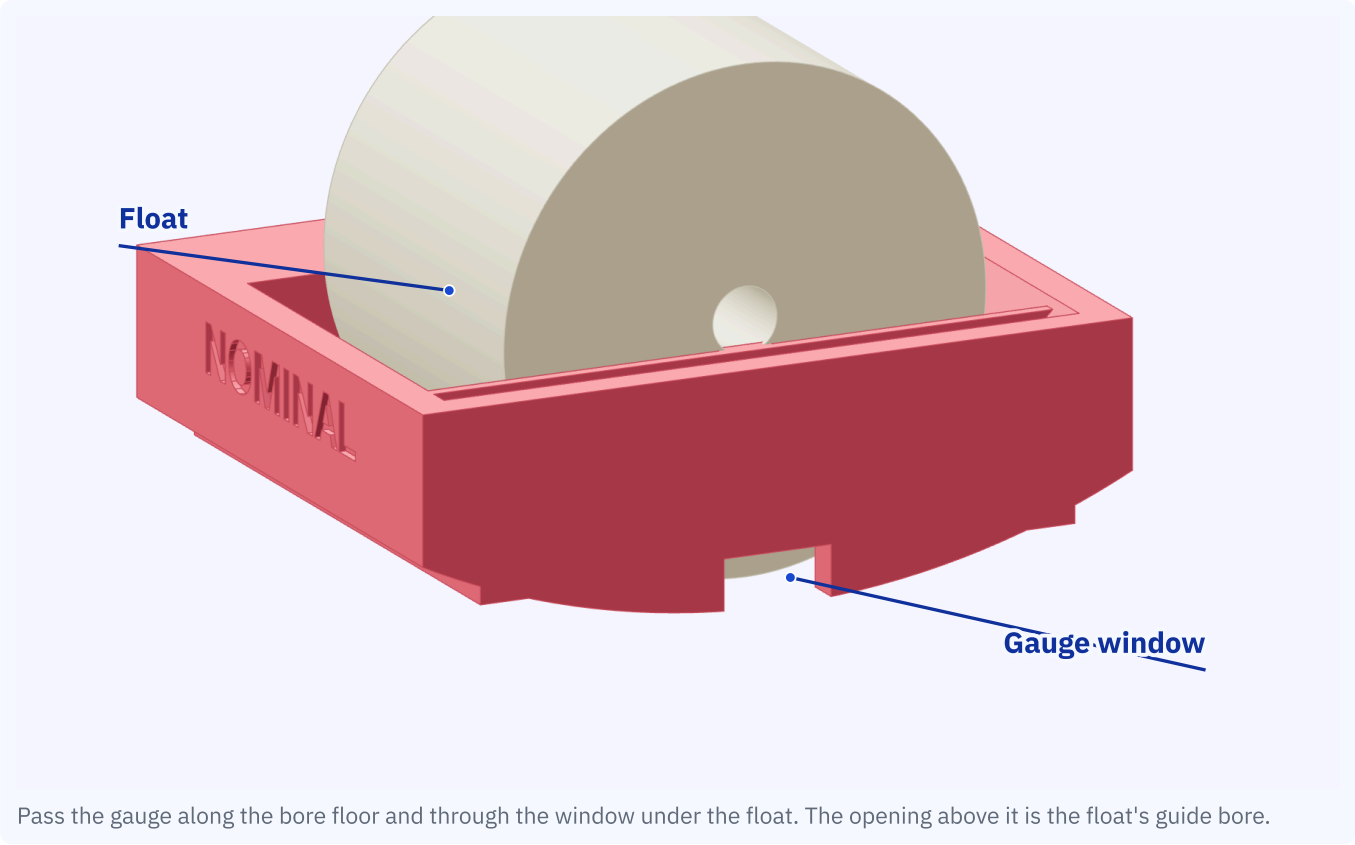
CHECK THE COUPLING

The roller stays seated through reversal. The sled rides on both runners and the float clears the tube. Stop at any bind and check the rod height, cradle seating and the sled's orientation.

SET THE FLOAT AND HOME

Read the float's wall gap

The gauge enters the window in the back of the sled from the open end away from the slide.



Pass the gauge along the bore floor and through the window under the float. The opening above it is the float's guide bore.

1. Try smooth gauge pins or smooth drill shanks. Find the largest that passes under the float without lifting it or rocking the sled.
2. Record the measured gap with the sled name. Compare with that sled's `float_to_wall_gap_mm` in the rig datums.
3. With the sled out, measure the slot width along the travel axis and subtract the roller diameter. The sled trails by that amount as the rod pushes it into the tube.

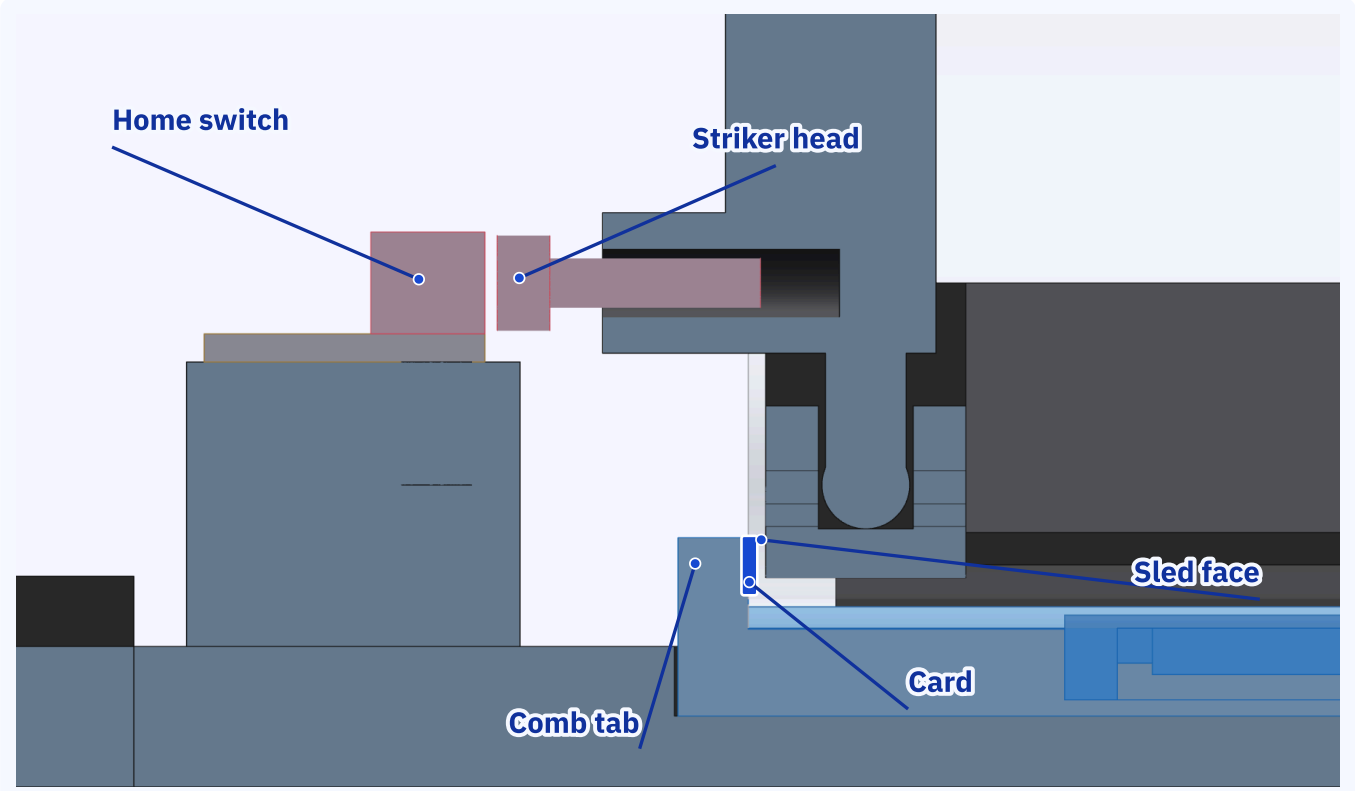
CHECK THE RUNNING GAP

The gauge enters without force. Remove it after recording the reading. The float must clear the bore in each sled used for measurement.

SET THE FLOAT AND HOME

Set the striker at the tab

The tab locates the tube rim. Set the switch to trip before the sled's leading face reaches that same stop.



Side cut through the mechanism near home. The blue card sets the space between the sled and the tab. Listen for the lever on the actual switch.

1. Leave the motor unpowered. Hold a bank card between the sled's leading face and the comb tab. Turn the coupling by hand until the sled just grips the card.
2. Back the striker screw out toward the switch until the lever clicks. Draw the card out.
3. Move the sled in until the switch releases, then bring it back slowly. Stop as soon as it clicks.

CHECK THE GAP AT THE CLICK

There is a card's thickness between the sled and the tab at the click. The switch releases as the sled moves into the tube and clicks at the same point on return.

WIRE THE CONTROLLER

Match the driver to the carrier

Identify the enable and motor-supply pins on both boards before fitting the TMC2209.



Use the pin names and confirm the orientation against the carrier in hand. This drawing shows the identifying ends of the two pin rows.

1. Disconnect USB and the motor adapter. Locate EN and VS on the driver and the matching socket positions on the carrier.
2. Set the carrier's three configuration switches to OFF for standalone STEP/DIR. The TMC2209 uses eight microsteps when both MS inputs are low.
3. With the driver still out, confirm both socket rows match the driver's pinout. Leave it out for the wiring and carrier meter checks.

CHECK THE SOCKET

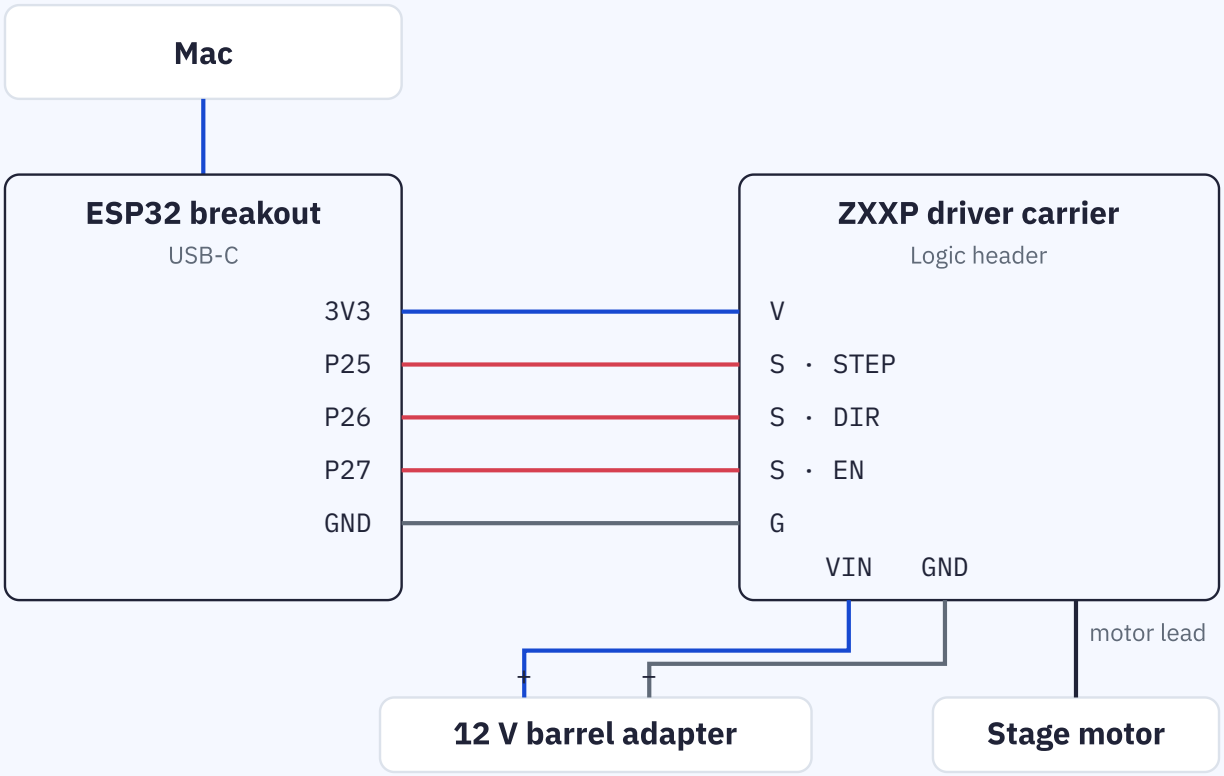
EN, STEP and DIR go to their labeled signals. Logic supply goes to VCC_IO, motor supply to VS, and ground to GND. Every pin fits its own socket with both rows aligned.

BIGTREETECH pinout and current adjustment

WIRE THE CONTROLLER

Wire the drive and supplies

The Mac powers the ESP32 over USB-C. The barrel adapter supplies motor power at the driver carrier.



On the carrier, S is the signal row. V feeds the driver's logic, G is its return, and VIN takes motor power. The sensor branches are on the next page.

1. With both supplies unplugged, match the breakout terminals to the carrier labels. Wire the barrel connector's marked plus to VIN and minus to the carrier's power GND.
2. At the loose motor plug, use resistance to identify the two coil pairs. The expected pairs are black/green and blue/red. Each pair must land on its own phase, 1A/1B or 2A/2B.

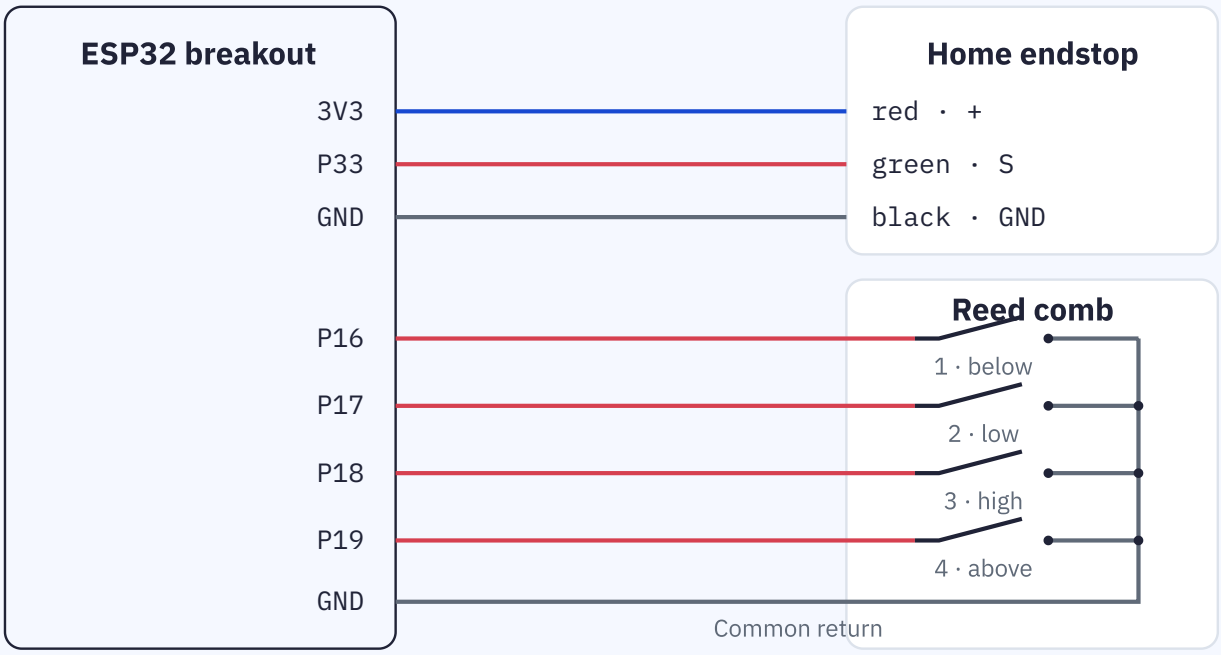
LEAVE POWER DISCONNECTED

Check each wire against the diagram. Leave the motor plug out through the current adjustment.

WIRE THE CONTROLLER

Wire the switch and reeds

Each reed closes its own input to the common return. The home switch has a separate signal and takes power from the breakout's 3V3 terminal.



The ESP32 pulls each reed input up. A closed contact pulls its station input to ground.

1. Match the endstop lead colors against the board's markings. Run the red supply to 3V3, black return to GND and green signal to P33.
2. Land the numbered comb signals in order from P16 through P19. Land the comb's common return at GND.
3. Route leads outside the cradle and away from the rod, carriage and coupling.

CHECK STATION ORDER

Trace each pocket back to its terminal from the tab end. The home lead and comb wires stay clear throughout the hand sweep.

CHECK THE WIRING

Meter the wiring before power

Pull USB and the barrel adapter. Leave the driver and motor unplugged while you trace the carrier and harness.

PROBE AT	CHECK AT THE OTHER END
Barrel connector	Plus reaches carrier VIN. Minus reaches carrier power GND and the logic return.
Breakout 3V3	Reaches carrier V and endstop supply. On the empty carrier, V reaches the driver's VCC_IO socket.
Breakout GND	Reaches carrier G, endstop ground and reed return.
P25 / P26 / P27	Reach STEP / DIR / EN on the carrier and their respective driver sockets.
P33 and reed inputs	Reach the switch signal and the matching comb leads.
Unplugged motor	Each phase has finite resistance within its pair. A wire from one pair to the other reads open.

CHECK FOR BRIDGES

VIN to GND, VIN to 3V3, and 3V3 to GND must show no persistent short. Check for strands bridging neighboring terminals and tug each connection lightly.

PROVE THE CARRIER'S LOGIC RAIL

Disconnect the whole logic header and leave the driver out. Apply the adapter to the carrier alone. Meter VIN to GND for the correct positive supply and V to G for an undriven logic rail. If the carrier sources voltage at V, stop and check its circuit before joining it to the ESP32's 3V3. Unplug the adapter and reconnect the logic header.

CHECK THE WIRING

Prove the inputs on USB

Leave the adapter and motor unplugged. Use the ESP32 console to check the reeds and home switch before the carriage can move.

Load the rig controller

Find the board's port with `hsm-peripheral boards`. Run the commands below from the checkout. The monitor reads its baud rate from the rig's PlatformIO target.

Watch the reed inputs

Lift the roller free and move the loaded sled along the tube by hand. Each wired station reports an `R` line on closure and release. Put the roller back in its slot once the check is done.

```
pio run -e reed_rig -t upload --upload-port <port>
pio device monitor -e reed_rig --port <port> --echo
```

Watch the home input

1. Send `status` with the lever free. Press and hold it, then send `status` again. The console should read `home free`, then `home pressed`.
2. If those states are reversed, set `homelevel low` for a pin that goes low when pressed or `homelevel high` for one that goes high.
3. Repeat the press and release after any polarity change or controller reset. Leave the lever free.

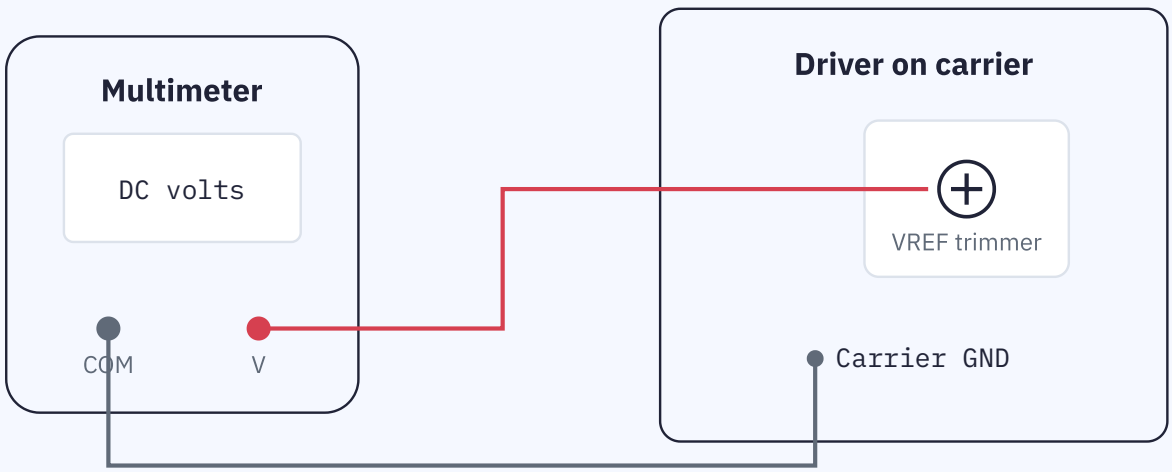
BOTH INPUTS ARE PROVEN

Each intended reed changes on its assigned station and the home input follows the real lever. The endstop's light is a local indication. Use the console reading to confirm what the ESP32 sees.

CHECK THE WIRING

Set the drive current

Start at the low reference voltage specified in the rig procedure. The motor stays unplugged while you measure.



Clip COM to carrier ground before applying power. Touch the other probe to the driver's VREF point. Keep both probes clear of adjacent pins.

1. Disconnect both supplies. Seat the driver as on page 15 and reconnect the logic header. Leave the motor lead out.
2. Connect USB first, then the adapter. Measure VREF and adjust in small increments with an insulated tool to about half a volt. Read the meter after each adjustment.
3. Unplug the adapter and USB before plugging in the motor lead.

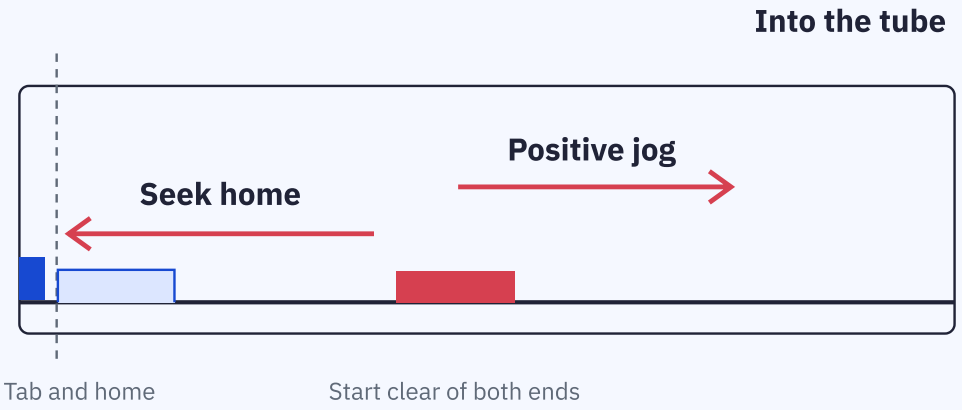
READY FOR A SHORT JOG

The initial reference is measured and recorded. Recheck the carrier and switch if any reading disagrees with the schematic. Keep the motor adapter within reach for first motion.

FIRST MOTION

Prove direction and home

Start with the loaded sled partway down the tube, clear of both ends, with the roller engaged. Keep the motor adapter within reach.



Positive is deeper into the tube. Home is the switch trip, with clearance left at the tab.

1. Connect USB and then the motor adapter. Repeat the free/pressed check if the ESP32 has restarted.
2. Send `jog 5`. The sled should move into the tube. If it moves toward the tab, stop and flip the saved direction with `dirinvert on` or `dirinvert off`, then jog again.
3. Send `home`. Watch it trip, back off, creep onto the switch and back off to release. Use the card's measured thickness at the trip for `tab <mm>`. The final stopped position includes the release travel.
4. Home again and look for the same stop. Compare a short commanded move with measured carriage travel.

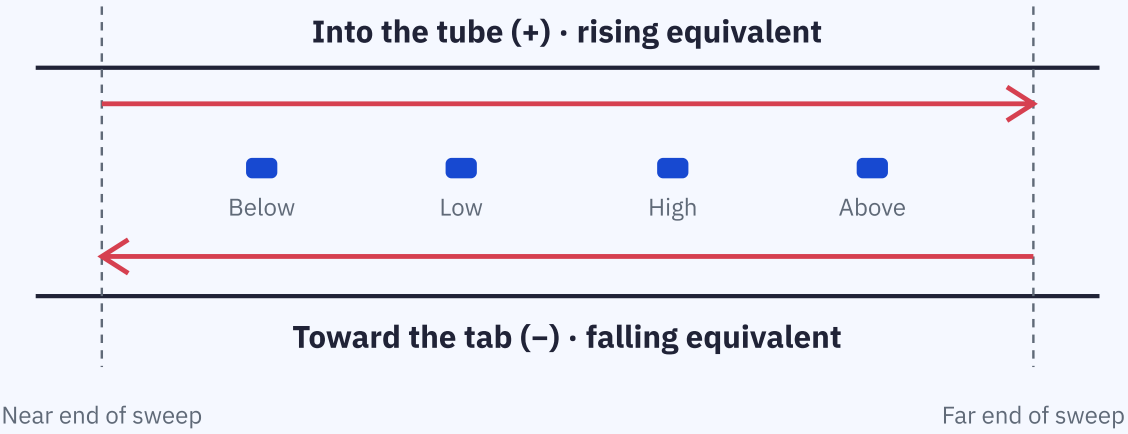
READY TO SWEEP

The switch trips before contact, the drive travels the commanded distance and nothing binds. Send a non-empty line to stop motion if the rig misbehaves. Home again before taking data after an interrupted move.

FIRST MOTION

Take a complete sweep

A sweep travels into the tube and returns at the same speed. Keep the direction with every contact change.



The diagram shows the sequence. Choose endpoints where the fitted reeds are open, within the reach reported by the controller.

```
hsm-peripheral reed-sweep --port <port> --sled nominal \
  --reeds 2=A,3=B --play <mm> --glass <mm> \
  --output <new-run-directory>
```

1. Clear loose metal. In the console, try one slow `sweep <from> <to>` inside the range from `status`. Both distances locate the sled's leading face from the tab.
2. Close the console. Run the host command with your port, sled and reed names. Pass the measured roller play as `--play` and the reed's glass length as `--glass`, both in millimetres.
3. The host homes and repeats the sweep. Press `Ctrl–C` to stop a host run. Keep the float and tube identities and the measured wall gap with `raw.log` and `sweep.json`.

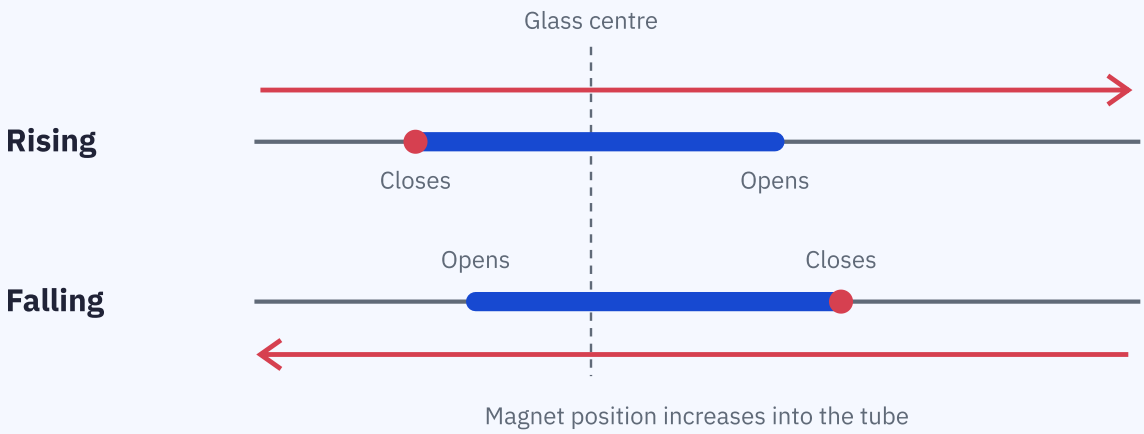
A COMPLETE RECORD

Each leg starts and ends in the log. For a clipped window, adjust `--from` or `--to` within proven travel. Resolve clipped or aborted runs before using their crossings.

CALIBRATION

Use the directional crossings

The dry rig measures the magnet's position relative to the reed. The float procedure supplies the waterline reading and the final liquid check.



The blue span is a closed contact. The sketch shows how to choose the closure for each direction. Use the positions from your run.

1. Capture NOMINAL, then NEAR and FAR, in separate directories. Home again after changing sleds.
2. Read the directional closing and opening positions and their spread in `sweep.json`. Keep separate activation regions and retain `raw.log` so the run can be replayed.
3. Use the inward closure for full and the return closure for empty. In the same height frame, add the measured magnet-to-liquid offset to the target liquid height, then subtract the signed magnet-to-reed offset at closure. Record the result in the reed calibration record.

CHECK IN LIQUID BEFORE FIXING THE REEDS

Confirm both control crossings are accessible and repeatable in the intended vessel. Apply the acceptance limits in the float calibration procedure before soldering fixed columns or printing the final bridge.

Rig procedure and log processing